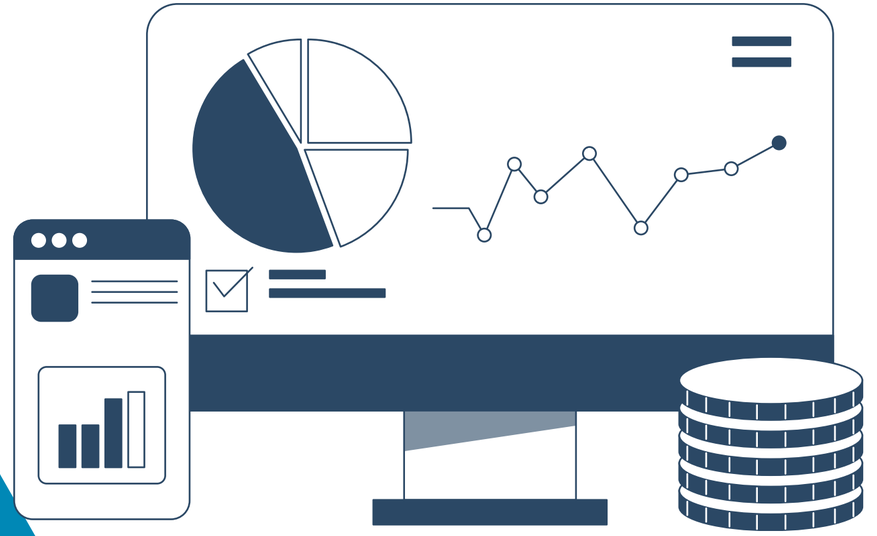
DATA ANALYTICS

30 days



DAY 20

Agenda

- Understand Window Functions
- Use ROW_NUMBER()
- Use RANK()
- Compare RANK vs ROW_NUMBER
- Find Top N Records
- Perform Advanced Business Analysis

What is a Window Function?

A Window Function performs calculations across a set of rows while keeping each row visible

Why Do We Need Window Functions?

Business Questions:

- Who is the top-selling employee?
- Which product ranks first?
- Top 3 customers?
- Highest revenue courses?

Create Table

```
CREATE TABLE SalesPerformance(
EmployeeID INT,
EmployeeName VARCHAR(50),
Revenue INT
);
```

EmployeeID	EmployeeNam	Revenue
1	Rahim	50000
2	Karim	70000
3	Nila	70000
4	Jony	40000
5	Sadia	60000

```
INSERT INTO SalesPerformance
VALUES
(1,'Rahim',50000),
(2,'Karim',70000),
(3,'Nila',70000),
(4,'Jony',40000),
(5,'Sadia',60000);
```

What is ROW_NUMBER()?

ROW_NUMBER() assigns a unique sequential number to each row.

Even if values are the same, numbers remain unique.

```
ROW_NUMBER()
OVER(ORDER BY ColumnName)
```

Employee	Revenue	RowNum
Karim	70000	1
Nila	70000	2
Sadia	60000	3
Rahim	50000	4
Jony	40000	5

```
SELECT
EmployeeName,
Revenue,
ROW_NUMBER()
OVER(ORDER BY Revenue DESC) AS RowNum
FROM SalesPerformance;
```

What is RANK()?

RANK() assigns rankings based on values.

Same values receive the same rank.

```
RANK()
OVER(ORDER BY ColumnName DESC)
```

Employee	Revenue	Rank
Karim	70000	1
Nila	70000	1
Sadia	60000	3
Rahim	50000	4
Jony	40000	5

```
SELECT
EmployeeName,
Revenue,
RANK()
OVER(ORDER BY Revenue DESC) AS Ranking
FROM SalesPerformance;
```

ROW_NUMBER vs RANK

Feature	ROW_NUMBER()	RANK()
Duplicate Values	Different Numbers	Same Rank
Skips Rank	No	Yes
Unique Sequence	Yes	No

Top 3 Employees

```
SELECT *
FROM (
SELECT
EmployeeName,
Revenue,
ROW_NUMBER()
OVER(ORDER BY Revenue DESC) AS RowNum
FROM SalesPerformance
) AS T
WHERE RowNum <= 3;
```

PARTITION BY Example

```
SELECT
EmployeeName,
Region,
Revenue,
RANK()
OVER(
PARTITION BY Region
ORDER BY Revenue DESC
) AS RankInRegion
FROM EmployeeSales;
```

Employee	Region	Revenue	Rank
Karim	Dhaka	70000	1
Rahim	Dhaka	50000	2
Nila	Khulna	60000	1
Jony	Khulna	40000	2

Common Business Use Cases

- Top Customers
- Top Products
- Top Courses
- Top Regions
- Sales Leaderboard
- Employee Performance Analysis
- Customer Segmentation

Assignment

Create a table named **CourseRevenue** with at least 5 courses and their revenues. Use **ROW_NUMBER()** to assign a unique sequence number and **RANK()** to rank the courses based on revenue. Identify the top 3 highest revenue-generating courses and submit your SQL queries with output screenshots.

THANK YOU

FOR YOUR ATTENTION